

AIMO Proof Pilot — Final Grade

Problem (#_ID): 3_INDEX

Submission: model_1

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The opening setup is valid, but the solution then goes wrong and there is nothing to credit. It asserts that index 4 is impossible without proof (and the claim is in fact false), never considers indices 5–9, and concludes that the maximum index is $N = 3$ — which is incorrect (the true value is 5). The subsequent optimisation of M and of the minimum size is therefore carried out over the wrong class of multisets, and rests on experiments over intervals $\{1, \dots, M\}$ rather than proofs. Since the submission contains no valid proof that the index is ≤ 6 , it does not even reach the one-mark partial-progress milestone. Scores **0**.